

Savitzky-Golay window ablation numerical diagnostics

No new plots are generated by this diagnostic summary.

W=0.0528807640069 s

acceleration-residual dynamics loss=846.8707362849622
selected rotor delay=0.190004539397 s
selected gimbal delay=0.021492752532 s
optimizer status=1 success=True nfev=6 optimality=1.9842485698973178e-07
final gradient inf-norm=1.9842485698973178e-07
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=429.1118502627755
finite-difference Jacobian max relative error=4.050542697931485e-12
confidence residual sample count=711

W=0.5 s

acceleration-residual dynamics loss=1.1006756753029818
selected rotor delay=0.2 s
selected gimbal delay=0.038669427119 s
optimizer status=1 success=True nfev=4 optimality=5.663680902046053e-08
final gradient inf-norm=5.663680902046053e-08
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=408.250079270499
finite-difference Jacobian max relative error=9.312112664126403e-12
confidence residual sample count=658

W=1 s

acceleration-residual dynamics loss=1.0830162381987127
selected rotor delay=0.195002269745 s
selected gimbal delay=0.036393417803 s
optimizer status=1 success=True nfev=4 optimality=1.412937660205671e-07
final gradient inf-norm=1.412937660205671e-07
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=405.0893028901442
finite-difference Jacobian max relative error=7.33331540232405e-12
confidence residual sample count=598

W=1.5 s

acceleration-residual dynamics loss=1.143496533398521
selected rotor delay=0.195393363886 s
selected gimbal delay=0.039327304411 s
optimizer status=1 success=True nfev=4 optimality=5.775235756289021e-08
final gradient inf-norm=5.775235756289021e-08
gradient tolerance satisfied=True
Jacobian rank=14 cond(nonzero)=410.4150611665243
finite-difference Jacobian max relative error=8.195483990518276e-12
confidence residual sample count=538

W=2 s

acceleration-residual dynamics loss=1.1574630856616104
selected rotor delay=0.2 s
selected gimbal delay=0.032269016751 s
optimizer status=1 success=True nfev=4 optimality=8.29991506634542e-08
final gradient inf-norm=8.29991506634542e-08
gradient tolerance satisfied=True

Jacobian rank=14 cond(nonzero)=390.70662516062816
finite-difference Jacobian max relative error=7.599270019489056e-12
confidence residual sample count=478